

Overview

It compresses data into an FP-Tree and extracts patterns directly from it.

How it works

1. Data Compression: compresses data into an FP-Tree
2. Mining the Tree: identify patterns that appear frequently
3. Generating Patterns: generates the frequent patterns (itemsets) avnd the rules

Construction of FP-Tree

Dataset: Coffee Shop Transactions Dataset for Market Basket

1. Compute the frequency of each individual item
2. Sort the item set in descending order (for simplicity, only the first 5 transactions are shown in the following examples)

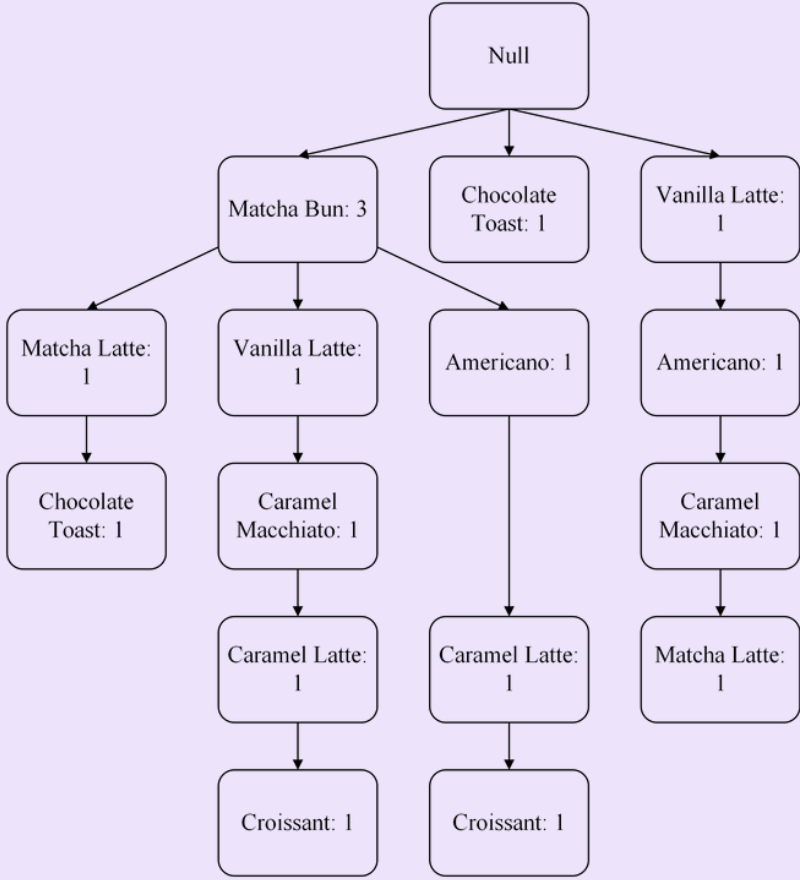
Items	Frequency	Vanilla Latte	2
Matcha Bun	3	Creamy Aren Latte	1
Croissant	2	Crispy Chicken Nuggets	1
Caramel Macchiato	2	Caramel Latte	2
Chocolate Toast	2	Butter Toast	1
Americano	2	Cookies	1
Matcha Latte	2	Latte	1

3. Remove items less than the minimum support (let minimum support be 2)
Frequent item set, L = {Matcha Bun: 3, Vanilla Latte: 2,Americano: 2, Caramel Macchiato: 2, Matcha Latte: 2, Caramel Latte: 2, Croissant: 2, Chocolate Toast: 2}

4. Ordered-Item set

ID	Items	Ordered-Item Set
1	{Vanilla Latte, Caramel Latte, Caramel Macchiato, Matcha Bun, Croissant}	{Matcha Bun, Vanilla Latte, Caramel Macchiato, Caramel Latte, Croissant}
2	{Creamy Aren Latte, Chocolate Toast, Cookies, Crispy Chicken Nuggets}	{Chocolate Toast}
3	{Caramel Latte,Americano, Matcha Bun, Croissant}	{Matcha Bun,Americano, Caramel Latte, Croissant}
4	{Latte, Matcha Latte, Matcha Bun, Chocolate Toast}	{Matcha Bun, Matcha Latte, Chocolate Toast}
5	{Vanilla Latte, Matcha Latte, Caramel Macchiato,Americano, Butter Toast}	{Vanilla Latte,Americano, Caramel Macchiato, Matcha Latte}

5. Draw FP Tree



Benefits of FP Tree structure

Completeness: FP-tree compresses the dataset while preserving all information needed to mine frequent itemsets

Compactness: Transactions with the same item order and prefix are merged into single paths, making it efficient for large datasets.



Mining frequent patterns from FP-Tree

1. Conditional Pattern Base
2. Conditional Frequent Pattern Tree
3. Frequent Pattern rules are generated

Items	Conditional Pattern Base	Conditional Frequent Pattern Tree	Frequent Patterns Generated
Matcha Bun			
Vanilla Latte	{ {Matcha Bun: 1} }		
Americano	{ {Matcha Bun: 1}, {Vanilla Latte: 1} }		
Caramel Macchiato	{ {Matcha Bun, Vanilla Latte: 1}, {Vanilla Latte,Americano: 1} }		
Matcha Latte	{ {Matcha Latte: 1}, {Vanilla Latte,Americano, Caramel Macchiato: 1} }		
Caramel Latte	{ {Matcha Bun, Vanilla Latte, Caramel Macchiato: 1}, {Matcha Bun,Americano: 1} }	{Matcha Bun: 2}	{<Caramel Latte, Matcha Bun: 2>}
Croissant	{ {Matcha Bun, Vanilla Latte, Caramel Macchiato, Caramel Latte: 1}, {Matcha Bun,Americano, Caramel Latte: 1} }	{Matcha Bun: 2}	{<Croissant, Matcha Bun: 2>}
Chocolate Toast	{ {Matcha Bun, Matcha Latte: 1} }		

Advantages vs. Disadvantages of FP Growth Algorithm



Advantages of FP-Growth

- Avoids expensive candidate generation (unlike Apriori).
- Uses a compact FP-tree to store shared item prefixes.
- Scans the database only twice.
- Faster and more scalable for large, dense datasets.
- Great for discovering frequent patterns in co-occurring data.

Disadvantages of FP-Growth

- Complex to build with many unique or long transactions.
- High memory use when compression is low.
- Harder to implement and explain than simpler methods.
- Less effective for sparse or high-dimensional datasets.
- Compression benefit drops with low support thresholds.

Applications of Association Rules

Association rule mining is used across different industries to discover relationships between items in large datasets. They help in understanding customer behavior, improving services, and making strategic decisions.

In this example, we use association rules in retail to identify products frequently bought together. For example, if many customers who buy coffee also buy bread, the coffee shop might sell them as a set to attract more people to buy them.

From the frequent pattern generated, we noticed that {Caramel Latte, Matcha Bun} and {Croissant, Matcha Bun} appeared together frequently.

Known that Confidence (A -> B) = Support (A ∪ B) / Support (A),
Confidence (Matcha Bun-> Caramel Latte) = 2/3 = 66.7%

Confidence (Matcha Bun -> Croissant) = 2/3 = 66.7%

By calculating the confidence of (Matcha Bun-> Caramel Latte) and (Matcha Bun -> Croissant), we can say that if someone buys a Matcha Bun, there is a 66.7% chance they will buy a Caramel Latte, or Croissant in the same purchase. This insight can be applied for product placement or bundling offers in a café menu.

CHALLENGE

Challenges in Association Rules Mining

Combinatorial Explosion

Too many possible itemsets (exponential growth); requires efficient pruning to stay feasible.

Rare but Critical Patterns

Important but infrequent patterns (e.g., fraud, anomalies) often missed due to low support.

Dynamic & Streaming Data

Most algorithms assume static data; real-time systems need incremental or online mining methods.



Lam Yoke Yu
A23CS0233



Tan Yi Ya
A23CS0187



Tan Zhi Ming
A23CS0189